Lab 5 Rubric: Momentum Conservation

Out of 100 points. This is the rubric on the LB5-M4 assignment in Brightspace. Lab 5 page · Word copy · PDF copy

Use this rubric as your writing guide: the criteria follow the lab report template top to bottom. Full-credit descriptors say exactly what a complete answer contains, so read them before you write rather than after.

Grammar, spelling, and formatting problems are assessed 1 point per occurrence, capped at 3 points for the report. A fault already scored under a criterion is not deducted again as a formatting occurrence. Deadline: a report submitted by the deadline may be revised and resubmitted up to two more times. A report that arrives after the deadline but before grades are posted is graded as on time but cannot be revised; one that arrives after grades are posted is late, may take a penalty, and gets no detailed feedback.

The phrase “human error” earns no credit anywhere in the report. A percent difference under 10% may be attributed to ordinary experimental uncertainty, provided you say so explicitly and name a mechanism. Trial 2 is the exception to the percent-difference rule — see Part III of the handout; quoting its percentage without comment is treated as a misreading of the measurement.

Criterion	Pts	Complete	Strong	Developing	Needs work	Missing
Heading	3	**3** Names, course and section, lab title and date, all present and correctly formatted. No template placeholders.	**2** One item missing or misformatted.	**2** Template placeholders or brackets left in the heading block.	**1** Several items missing.	**0** No heading block.
Abstract	7	**7** Four sentences or fewer. States what was measured, gives the three Δp values with units, states whether momentum was conserved in each trial, and notes the trial where the percent difference could not be used. No methods or apparatus.	**6** All content present but runs long or includes method detail.	**4** Reports some results but omits the Δp values or the conservation verdict.	**2** Describes the lab but reports no numbers.	**0** No abstract.
Introduction	8	**8** States the conservation law in the student's own words and builds Σp_before = Σp_after as a sum over the bodies, with every symbol defined. Says what 'conserved' means as a measurable comparison here. Ends with a testable hypothesis.	**7** All elements present; one definition thin or the hypothesis weakly stated.	**5** The equation is quoted rather than built, or 'conserved' is left as a sentiment rather than a comparison.	**3** Law stated with no equation, or equation with no definitions.	**0** No introduction.
Experimental: Materials & Setup	8	**8** Apparatus described in prose; a labelled sketch showing the track, both carts and both sensors, with the positive direction marked; both cart masses and the extra bar's mass given. States the sign convention AND how it was established from the data rather than assumed.	**7** Sketch and masses present; the sign convention stated but not justified from the data.	**5** Sketch present but the positive direction is not marked, or a mass is missing.	**3** Bare equipment list, or no sketch.	**0** No materials or setup description.
Experimental: Procedure	8	**8** Written in own words as prose, reproducible. Says how the interaction was located in each run, how each fit window was chosen, and that the carts never touch and why that matters.	**7** Reproducible and in the student's own words; the fit-window reasoning is thin.	**5** Reproducible, but the window selection or the no-contact point is absent.	**3** Copied or closely paraphrased from the handout.	**0** No procedure.
Results: Data Tables	12	**12** Table of all twelve fits carrying trial, cart, phase, THE FIT WINDOW USED, and the signed velocity; plus a table of the three trials with Σp before, Σp after and Δp. Captions above, units in header rows, consistent significant figures.	**10** Both tables present; minor issues with captions, units or significant figures.	**7** Tables present but the fit window column is missing.	**4** Fewer than twelve fits tabulated, or velocities given without signs.	**0** No data tables.

Criterion	Pts	Complete	Strong	Developing	Needs work	Missing
Results: Graphs	**8**	**8** At least one run shown with both carts' position traces and the fitted lines drawn on, so the reader can see which points were kept and where the interaction was judged to be. Axes labelled with units; caption below.	**7** Figure present and correctly plotted; caption or axis labels incomplete.	**5** Traces shown but no fitted lines, so the window cannot be seen.	**3** Points joined dot-to-dot, or axes unlabelled.	**0** No graph.
Results: Sample Calculations	**13**	**13** One worked sample of each type, with units at every step and the numbers shown going in: a velocity from a fit, a Σp, a Δp, and a percent difference. Energy shown too if the energy question was chosen.	**11** All types present with minor unit or rounding slips.	**8** Two or three of the four types worked.	**4** Some types missing, or units given only in the final answer.	**0** No sample calculations.
Discussion	**15**	**15** Opens with the three Δp values and a plain conserved/not-conserved verdict for each. Addresses the 10% criterion by name for Trials 1 and 3. For Trial 2, explains why the percent difference is the wrong tool and states what was reported instead. Accounts for the DIRECTION of each discrepancy and tests a physical explanation against the data.	**12** Most of the above; one element thin, usually the direction of the discrepancy.	**9** Opens with results and interprets them, but the Trial 2 problem is noted without being explained.	**5** Restates the results without interpreting them, or quotes Trial 2's percentage uncritically.	**0** No discussion.
Additional Questions	**8**	**8** Three of the five chosen and named, each answered fully in a dedicated section, in paragraphs with the working shown and not restating the question.	**6** Three answered but one thinly, or the three chosen are not named.	**4** Answered inside another section rather than a dedicated one, or only two answered.	**2** Answered as a bare list, or fewer than two answered.	**0** Not answered.
Conclusion	**5**	**5** Three to four sentences: the numerical results, a definitive verdict per trial, and one concrete improvement.	**4** Present but missing the numbers or the verdict.	**3** Present with numbers but no definitive verdict.	**2** Present but vague, or a repeat of the discussion.	**0** No conclusion.
References	**5**	**5** At least one source beyond the textbook; Note 8 and the handout count. Author, title and date for each, consistently formatted.	**4** Multiple references but formatting incomplete.	**3** Textbook and handout only, properly formatted.	**2** Handout only.	**0** No references.
Total	**100**					

Overall: Complete 90+ · Nearly there 75+ · Needs work 50+ · Missing below 50.

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